

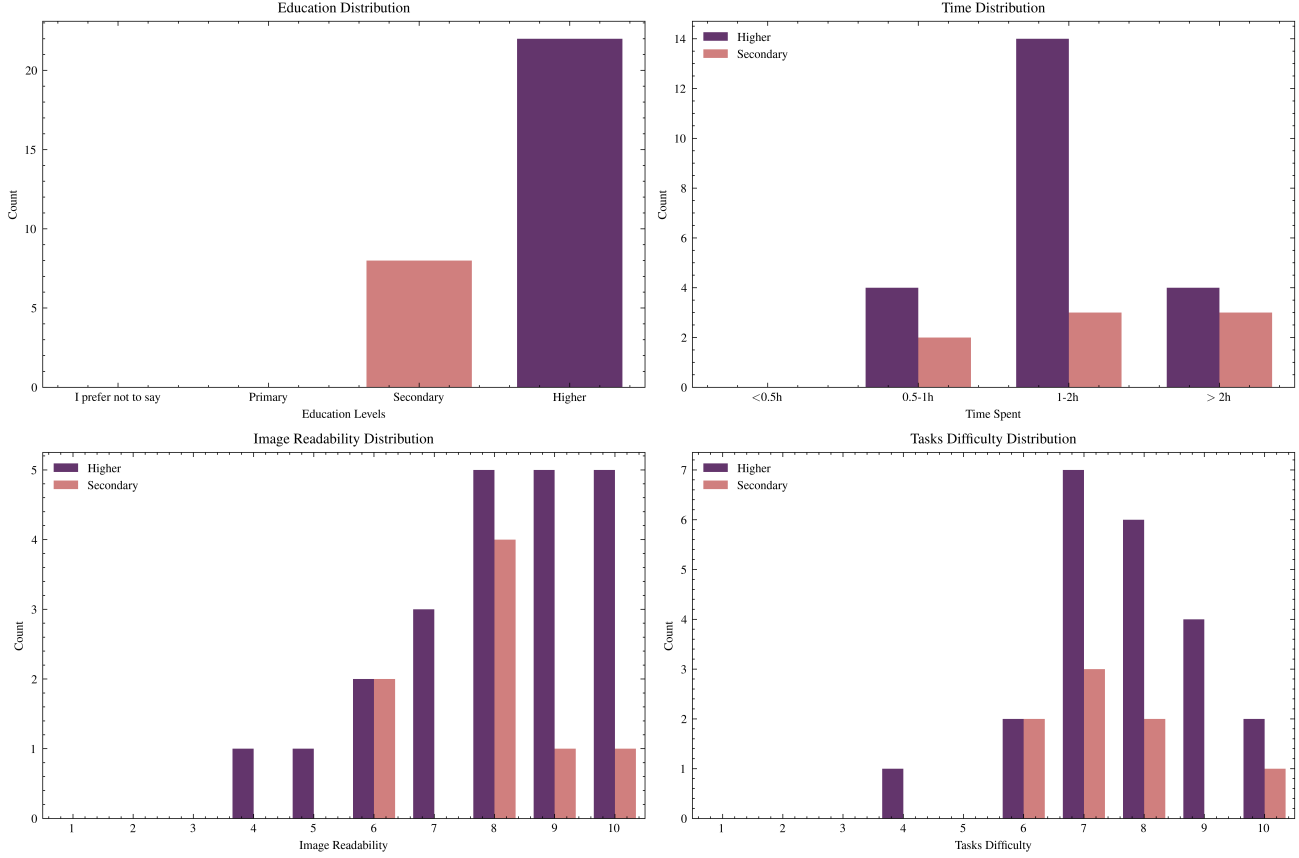


Figure 6: **The respondent’s answers to the questions attached in the form.** Even though the majority of respondents had higher education, they, on average, spent more than one hour on solving the problems. Moreover, only one person rated the problems as relatively easy (giving them a difficulty score of 4 out of 10).

demonstrate the superiority of humans over MLLMs in tackling this type of task. Notably, the lowest human score exceeded the combined score of all the models. Half of the respondents solved more than 40 problems, with mean and median equal to 39.2 and 40.5, resp., resulting in 65% average accuracy.

The difficulty of each problem can be estimated based on the number of respondents who successfully solved it. As shown in Fig. 8, the problems exhibit varying levels of difficulty: 22 of them were solved by at least 25 respondents, while 10 were solved by fewer than 10 respondents. In addition, three problems—10, 88 and 100—were solved by all respondents. Overall, the dataset was rated as quite difficult, with an average difficulty score of 7.6 across all respondents.

Overall, the results demonstrate the robustness and applicability of Bongard-RWR as a novel dataset for investigating the performance differences between human and model-based visual reasoning.

C. Extended Results

Table 4 presents results across all models, strategies and datasets discussed in Section 5. In the following paragraphs, we extend the discussion concerning binary and multi-class classification settings.

Binary classification (Ground-Truth). We assessed whether MLLMs can determine if a given concept matches a problem instance. On synthetic BPs, 3 proprietary (GPT-4o, Gemini 1.5 Pro, Claude 3.5 Sonnet) and 2 open-access (LLaVA-1.6 Mistral-7B, Pixtral 12B) models outperform a random classifier by a notable margin. On Bongard HOI, 3 proprietary (GPT-4o, GPT-4 Turbo, Gemini 1.5 Pro) and the same 2 open-access models also surpass random guessing. Notably, Pixtral 12B attained a perfect score on this dataset. On Bongard-OpenWorld GPT-4o, Gemini 1.5 Pro, LLaVA-1.6 Mistral-7B, and Pixtral 12B achieve reasonable results. Again, the leading model is Pixtral 12B with the outstanding 99% outcome. Model

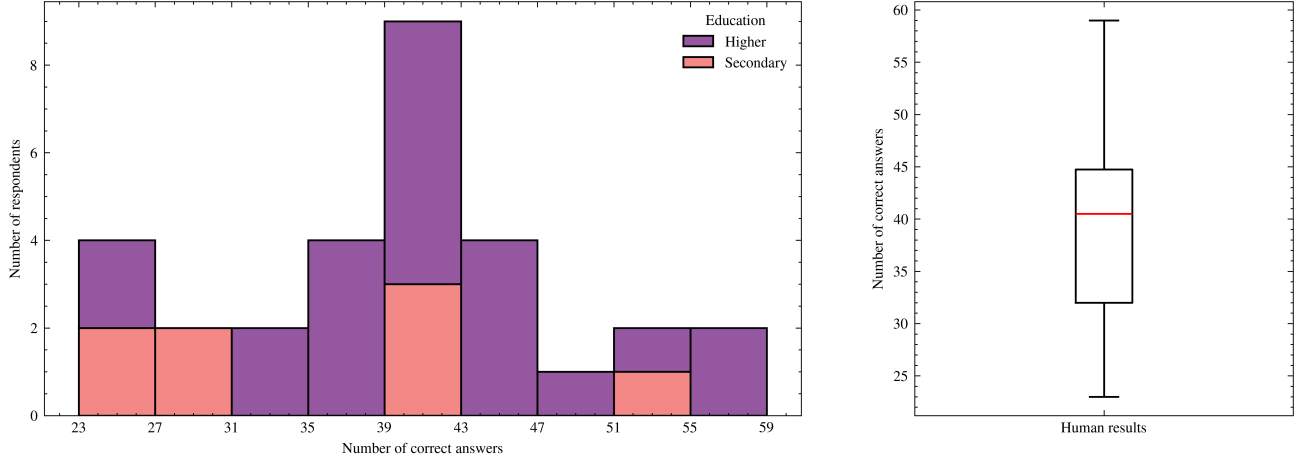


Figure 7: **The number of problems solved by humans.** The histogram illustrates the consistency across multiple respondents solving the Bongard-RWR problems. Notably, half of the respondents solved more than 40 problems, with none of them solving fewer than 23 ones. In the histogram, the lower bounds of the bins are inclusive, and the upper bounds are exclusive, except for the last bin, which is $[55, 59]$.

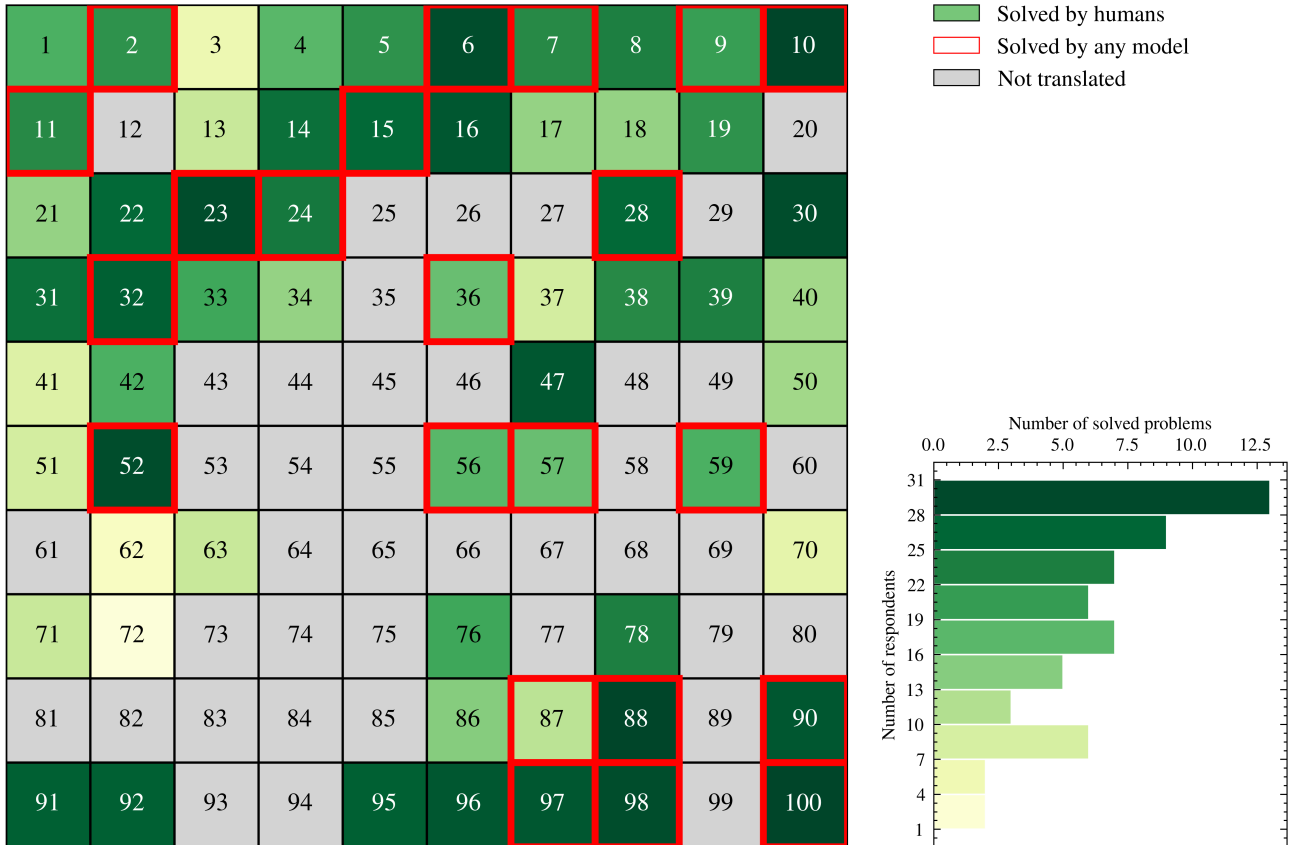


Figure 8: **Human performance on Bongard-RWR.** As shown in the plot, the models struggled with many problems that humans found relatively easy to solve. On the other hand, the models were able to solve problem #87 that appeared to be relatively demanding for human solvers. In the histogram, the lower bounds of the bins are inclusive, and the upper bounds are exclusive.

Table 2: **Binary classification on Bongard-RWR variants.** The number of problems solved by open-access models in the Images to Sides task from Bongard-RWR and its three variants, with the difference relative to Bongard-RWR in parentheses.

	BONGARD-RWR	RWR-G	RWR-S	RWR-SG
INTERNVL2 8B	16	17 (+ 1)	29 (+13)	27 (+11)
INTERNVL2.5 38B	13	13 (+ 0)	25 (+12)	23 (+10)
INTERNVL2.5 78B	15	17 (+ 2)	24 (+ 9)	23 (+ 8)
QWEN2VL 72B-INSTRUCT	20	25 (+ 5)	20 (+ 0)	21 (+ 1)
LLAVA-1.6 MISTRAL 7B	0	3 (+ 3)	0 (+ 0)	0 (+ 0)
LLAVA-NEXT 110B	24	29 (+ 5)	29 (+ 5)	32 (+ 8)
PHI-3.5-VISION	1	16 (+15)	8 (+ 7)	18 (+17)
PIXTRAL 12B	26	21 (− 5)	35 (+ 9)	33 (+ 7)

accuracy drops significantly on Bongard-RWR, where only LLaVA-1.6 Mistral-7B and Pixtral 12B outperform a random classifier. This suggests that correctly identifying concepts expressed in Bongard-RWR likely requires more advanced reasoning abilities, even in the relatively simpler binary classification setting.

Binary classification (Incorrect Label). Rejecting a possible solution is intuitively simpler than confirming its correctness, as it boils down to finding at least one image that doesn’t match the provided concept. Accordingly, 6 models perform better in the Incorrect Label setting than in Ground-Truth, with 7 perfect scores, 2 of them on Bongard-RWR. The exceptions are LLaVA-1.6 Mistral-7B, and Pixtral 12B which are below a random guessing threshold for all four datasets, despite being above this threshold in Ground-Truth. This suggests that their strong performance in the Ground-Truth setting may be due to a potential bias toward agreeing with the provided concept.

Binary classification (Images to Sides). We also evaluate the models’ ability to correctly assign two test images to the appropriate sides of the problem. A problem is considered solved if both images are correctly assigned to the respective sides. Proprietary models perform well in this task across synthetic BPs, Bongard HOI and Bongard-OpenWorld. Conversely, among open-access models, only Pixtral 12B consistently achieves strong results. Notably, on Bongard-RWR Pixtral 12B solves 26/60 problems, outperforming all proprietary models, however, all models perform below the level of random guessing. The remaining open-access models show poor results in this setting. Notably, weak results of LLaVA-1.6 Mistral-7B on real-world datasets are primarily attributed to its incorrect generation of JSON output required to format the result.

Binary classification on RWR variants. In the Ground-Truth and Incorrect Label settings, models performed similarly across the Bongard-RWR dataset and its three variants (RWR-G, RWR-S, RWR-SG), indicating that color removal and cropping did not substantially affect performance in these settings. However, in the Images to Sides setup (Table 2), several models showed clear differences. For example, Phi-3.5-Vision improved from solving 1 problem on RWR to 16 on RWR-G, 8 on RWR-S, and 18 on RWR-SG, suggesting that color removal and cropping reduce unnecessary variation, and shift the focus of the model toward structural patterns. Encouraged by these improvements, we evaluated larger model variants, including InternVL2.5 38B and 78B (Chen et al., 2024a;c), Qwen2VL 72B-Instruct (Bai et al., 2023; Wang et al., 2024), and LLaVA-NeXT 110B, which showed similar trends. Overall, the RWR-G variant highlighted models’ sensitivity to color, while RWR-S and RWR-SG revealed how cropping affects performance by reducing empty space between panels. Nevertheless, in most cases the models were unable to surpass the random guess threshold, highlighting their inherent limitations in recognizing Bongard-RWR concepts.

Multi-class classification (concept selection). Fig. 9 and Table 3 present detailed results of the multi-class classification experiments, where models were tasked with selecting the correct concept from $k \in \{2, 4, 8, 16\}$ choices across all four datasets. First, the performance improves from InternVL2 8B to InternVL2.5 8B, reflecting improvements in training data and model architecture in newer model versions. Next, moving from InternVL2.5 8B to InternVL2.5 38B and InternVL2.5 78B shows further gains, highlighting the benefits of scaling up model size, which we explore further in Appendix D. Comparing InternVL2.5 78B with Qwen2VL 72B-Instruct reveals that state-of-the-art models also from other families face challenges on this task, particularly on Bongard-RWR and synthetic BPs. It’s worth noting that the average human accuracy of 65% on Bongard-RWR observed in the free-form setup is similar to the performance of best models in the much simpler multiple-choice setup with only $k = 4$ candidate answers. This observation underscores the limitations of current models compared to humans.